KIROVAULT

Testing & Quality Assurance

Test strategy and acceptance criteria derived from the KiroVault blueprint. Targets are proposed test criteria, not claims of measured performance.

DOCUMENT PURPOSE

This standalone companion document complements the KiroVault Desktop App Blueprint. It follows the Windows-first, local-first product direction. Recommendations are proposed guidance, not claims that these features already exist.

Contents and implementation guidance

Test levels

• Unit: path validation, file-kind detection, date grouping, cache keys and conflict decisions.

• Integration: Rust commands, SQLite migrations, thumbnail providers and filesystem operations.

• UI: navigation, selection, preview routing, theme persistence and error states.

• End-to-end: browse, preview, move/copy/delete, scan/restart and disconnected-drive recovery.

• Release: installer, uninstaller, permissions, accessibility, licensing and clean-machine checks.

Acceptance criteria

• App starts without internet and becomes interactive before full-drive scanning.

• Folder names/metadata appear without waiting for every thumbnail.

• Grid/list/timeline, sorting, filters, favorites and navigation work predictably.

• Image previews show real content; compatible videos play; PDF and text/code previews work.

• Long folders remain responsive through virtualization and bounded preview work.

• Index can pause/cancel; stale paths reconcile; destructive actions require confirmation.

Test matrix

• Files: Unicode, long names, zero-byte, huge files, corrupted files and extension mismatch.

• Storage: C:, D:, removable drives, inaccessible folders, full disks and disconnected devices.

• Formats: images, mixed-codec videos, encrypted/damaged PDFs, office files, text/logs and nested ZIP.

• Operations: collisions, cross-drive copy, partial failures, cancelled operations and restart recovery.

• Accessibility: keyboard-only flow, contrast, scaling, reduced motion and long filenames.

Bug report template

• Environment: Windows version, app build, architecture and reproduction dataset.

• Steps to reproduce, expected result, actual result, frequency and severity.

• Attach redacted logs only; do not include private file content or paths unless necessary and approved.

Release gates

• All critical filesystem safety tests pass.

• No known data-loss defects; partial failures are visible.

• License inventory and third-party notices are complete.

• Install/uninstall and upgrade paths verified on a clean Windows environment.

References

• ISTQB glossary: https://glossary.istqb.org/

• Playwright documentation: https://playwright.dev/docs/intro

- Rust testing: https://doc.rust-lang.org/book/ch11-00-testing.html
- KiroVault Desktop App Blueprint, pp. 20–22 (user-provided source).

Reference note: external documentation may change. Verify current versions, licensing, APIs and platform prerequisites during implementation.